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Liu et al.

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(54) **TECHNIQUES FOR IMPLEMENTING
PHONE NUMBER-BASED USER ACCOUNTS
WITH PERMISSIONS TO ACCESS VARYING
LEVELS OF SERVICES UTILIZING VISIBLE
AND HIDDEN CONTACT ADDRESSES**

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Sep. 10, 2020, now Pat. No. 11,716,426.

(Continued)

(57) **ABSTRACT**

This application sets forth techniques for managing phone
number-based user accounts. According to some embodi-
ments, a server computing device can be configured to carry
out the steps of (1) receiving, from a first client computing
device, a request to create a user account based on a phone
number, (2) in response to authenticating that the first client
computing device is associated with the phone number:
creating the user account based on the phone number,
creating a hidden contact address based on the phone
number, and assigning the hidden contact address to the user
account, and (3) in response to receiving a notification that
a second client computing device is associated with the
phone number: requiring the user account to be updated to
be based on a different phone number, or assigning, to the
user account, a visible contact address based on the phone
number.

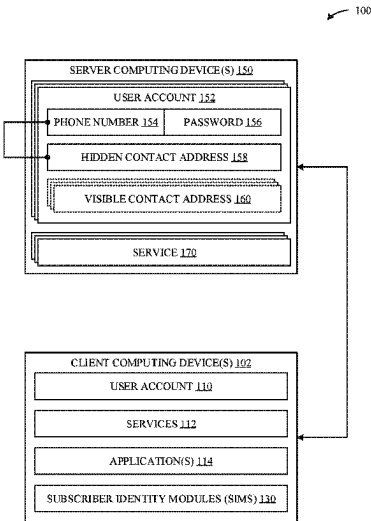
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H04M 15/00 (2024.01)
H04M 1/27457 (2020.01)
H04M 3/42 (2006.01)

(52) **U.S. Cl.**
CPC ... **H04M 15/7556** (2013.01); **H04M 1/27457**
(2020.01); **H04M 3/42382** (2013.01); **H04M**
2203/6081 (2013.01)

(58) **Field of Classification Search**
CPC H04M 15/7556; H04M 1/27457; H04M
3/42382; H04M 2203/6081; H04M 15/00;

(Continued)

20 Claims, 10 Drawing Sheets



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(58) **Field of Classification Search**

CPC H04M 15/8221; H04M 1/2745; H04L 63/102; H04L 67/306; H04L 63/0853; H04W 4/24; H04W 12/72

See application file for complete search history.

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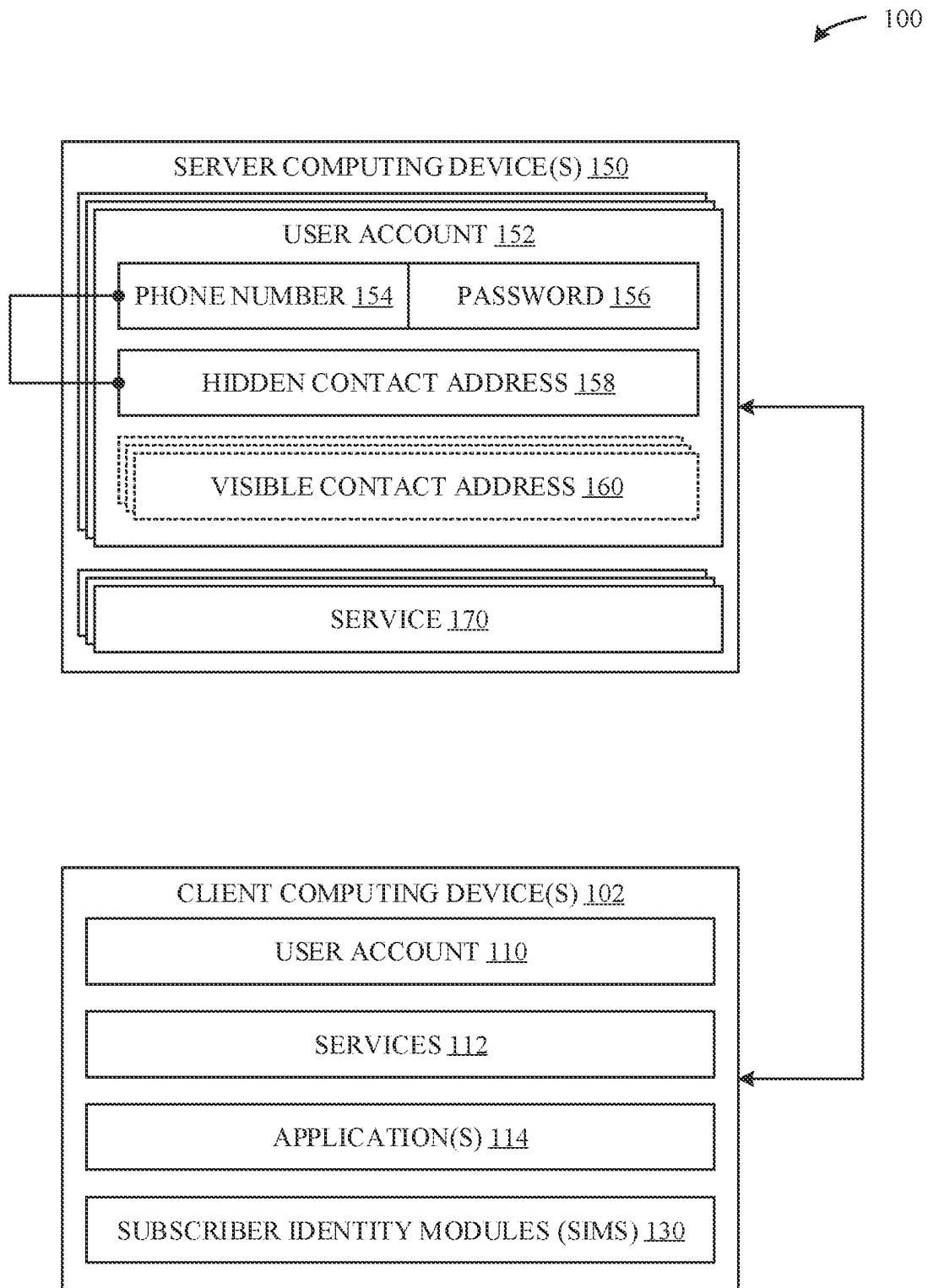


FIG. 1

↖ 200

Create New User Account

Phone Number: 111-222-3333

Password: XXXXXXXX

Verify Password: XXXXXXXX |

Submit

q w e r t y u i o p
a s d f g h j k l
↑ z x c v b n m ←
123 space return

FIG. 2A

210

Create New User Account

Phone Number: 111-222-3333

An SMS message with a code was sent to 111-222-3333.

Please type in the code here.

123456 |

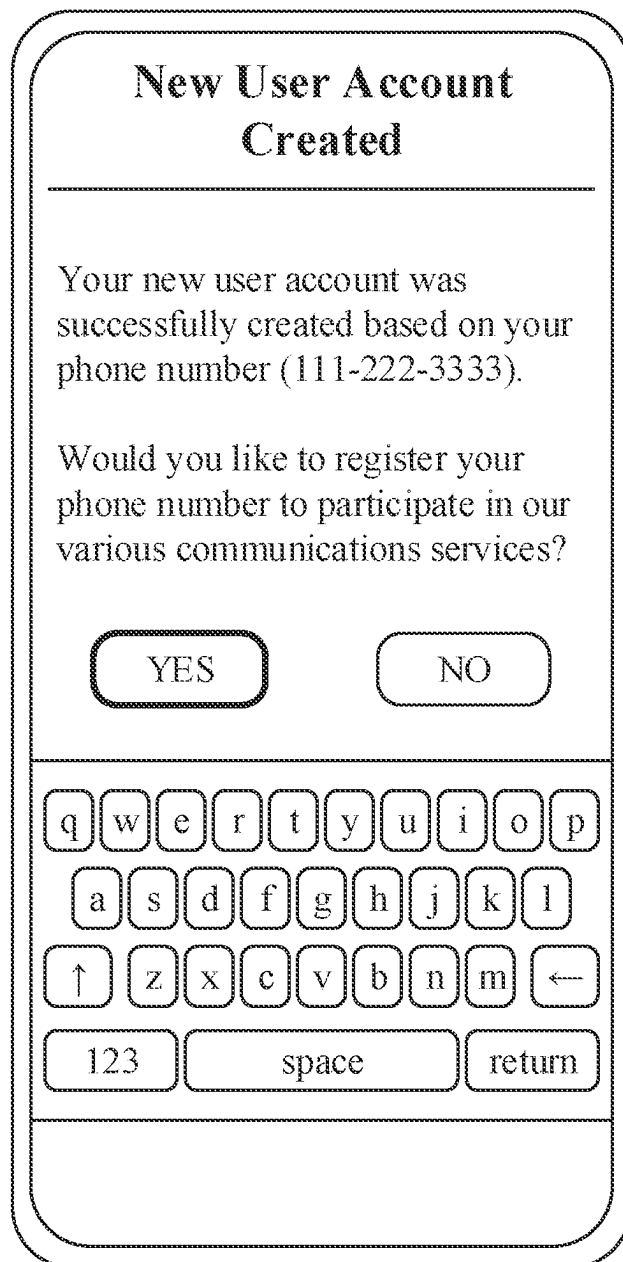
q w e r t y u i o p

a s d f g h j k l

↑ z x c v b n m ←

123 space return

FIG. 2B

 220

**New User Account
Created**

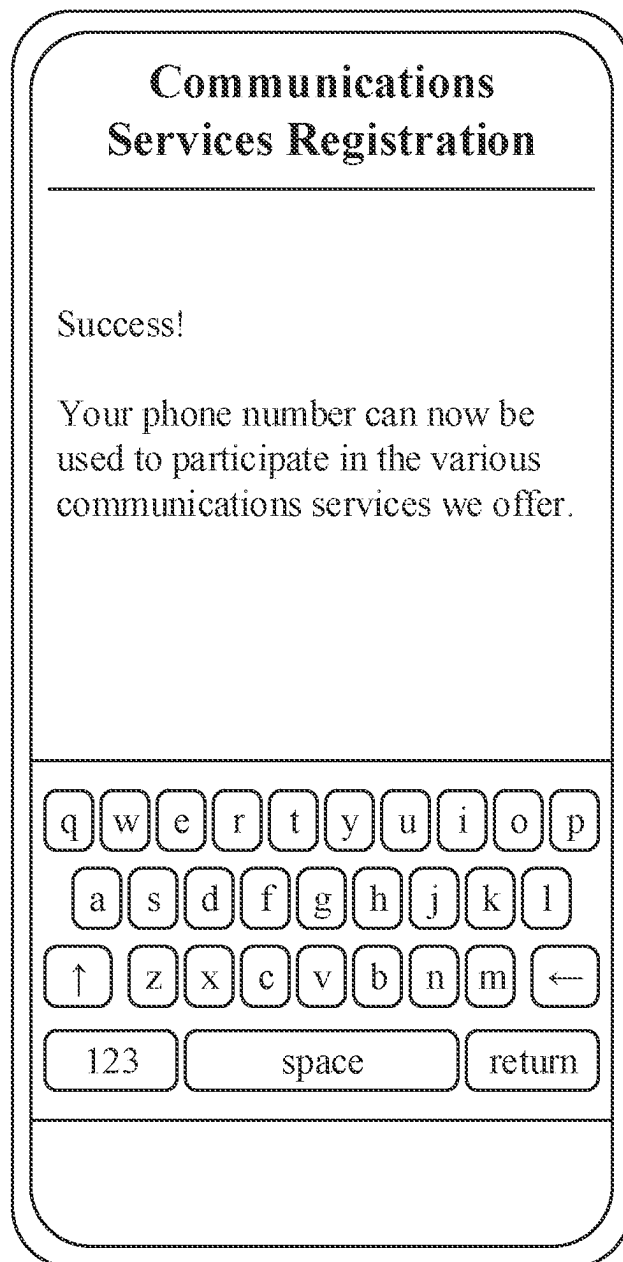
Your new user account was
successfully created based on your
phone number (111-222-3333).

Would you like to register your
phone number to participate in our
various communications services?

q	w	e	r	t	y	u	i	o	p
a	s	d	f	g	h	j	k	l	
↑	z	x	c	v	b	n	m	←	
123	space					return			

FIG. 2C

230

*FIG. 2D*

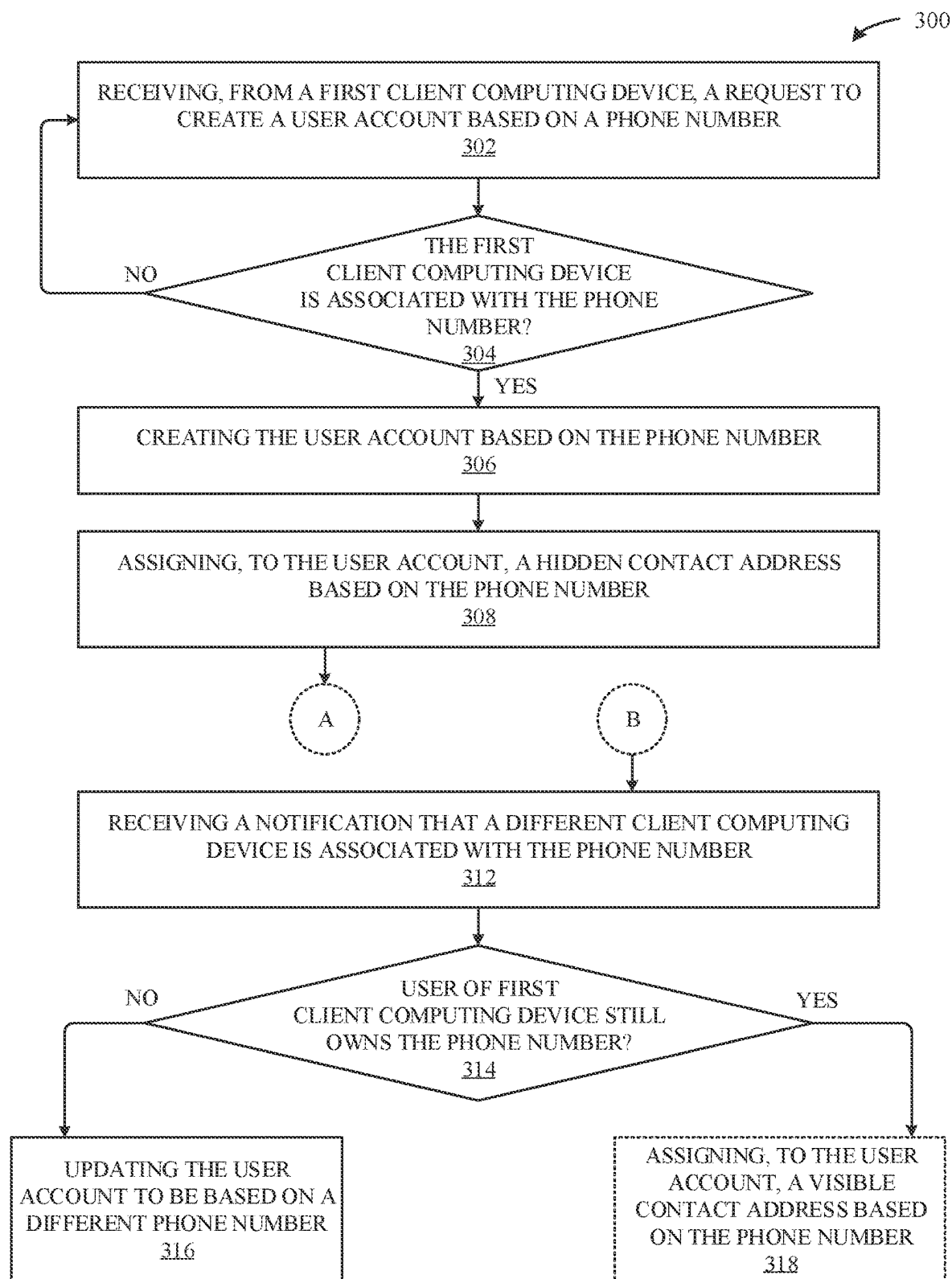


FIG. 3A

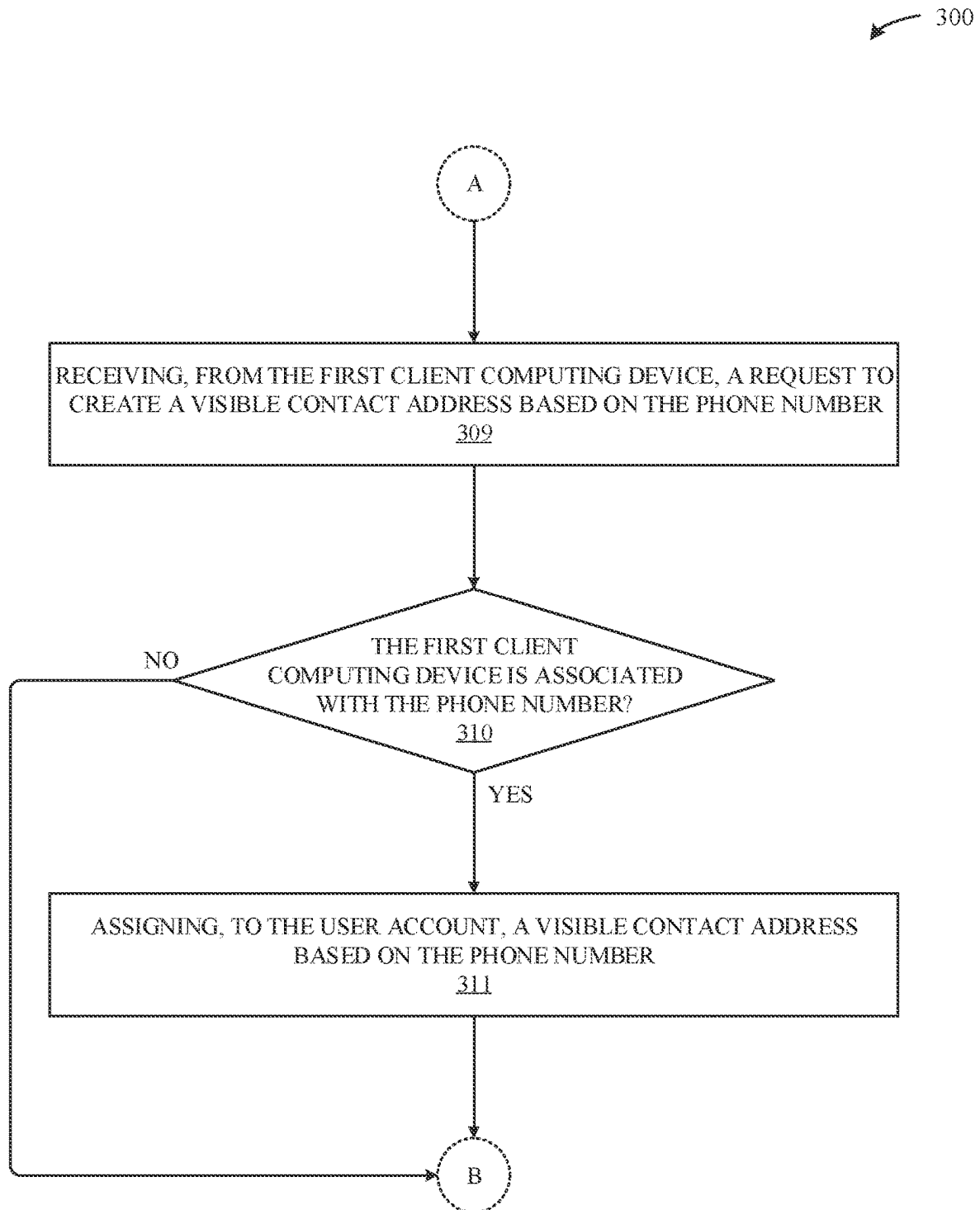


FIG. 3B

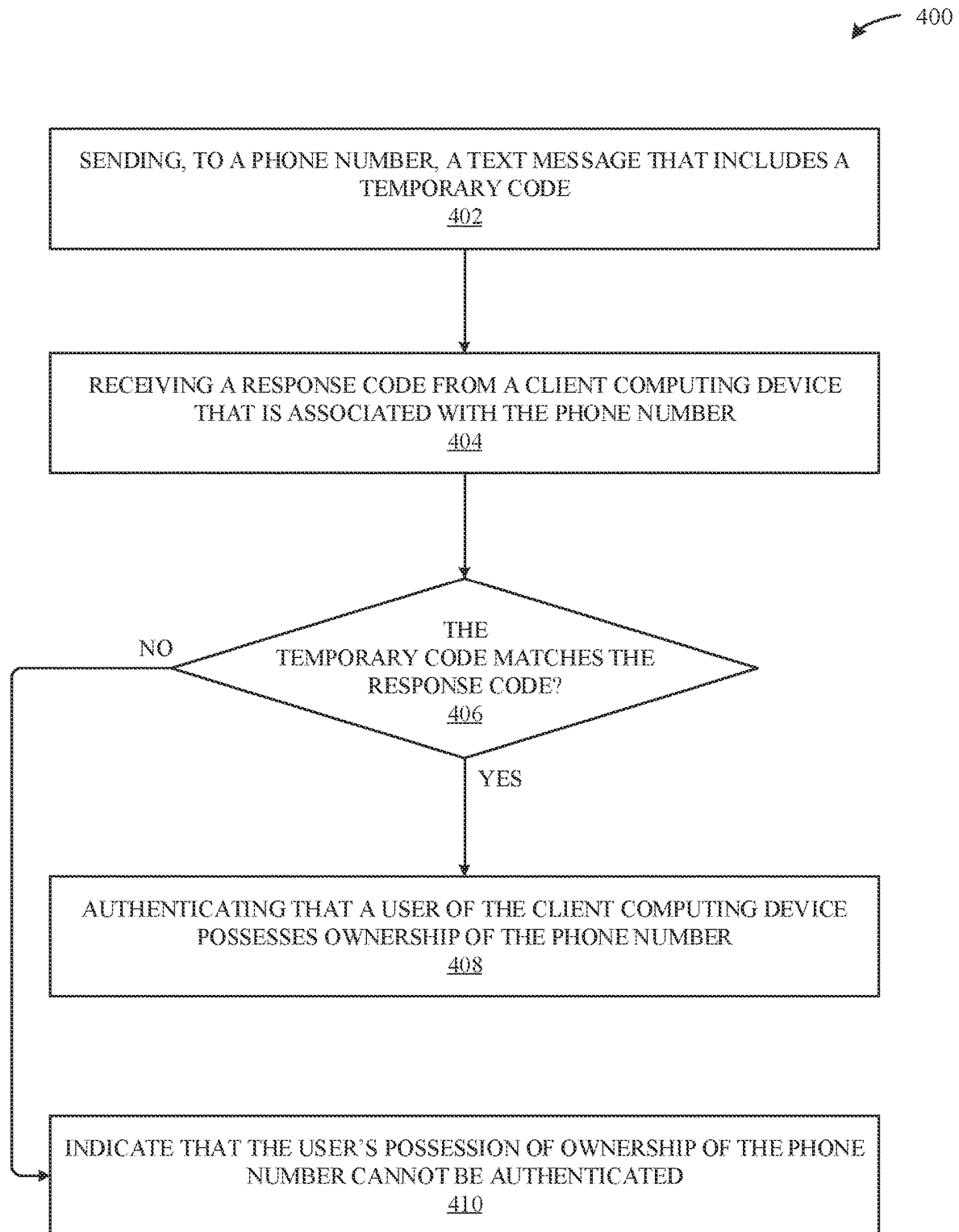


FIG. 4

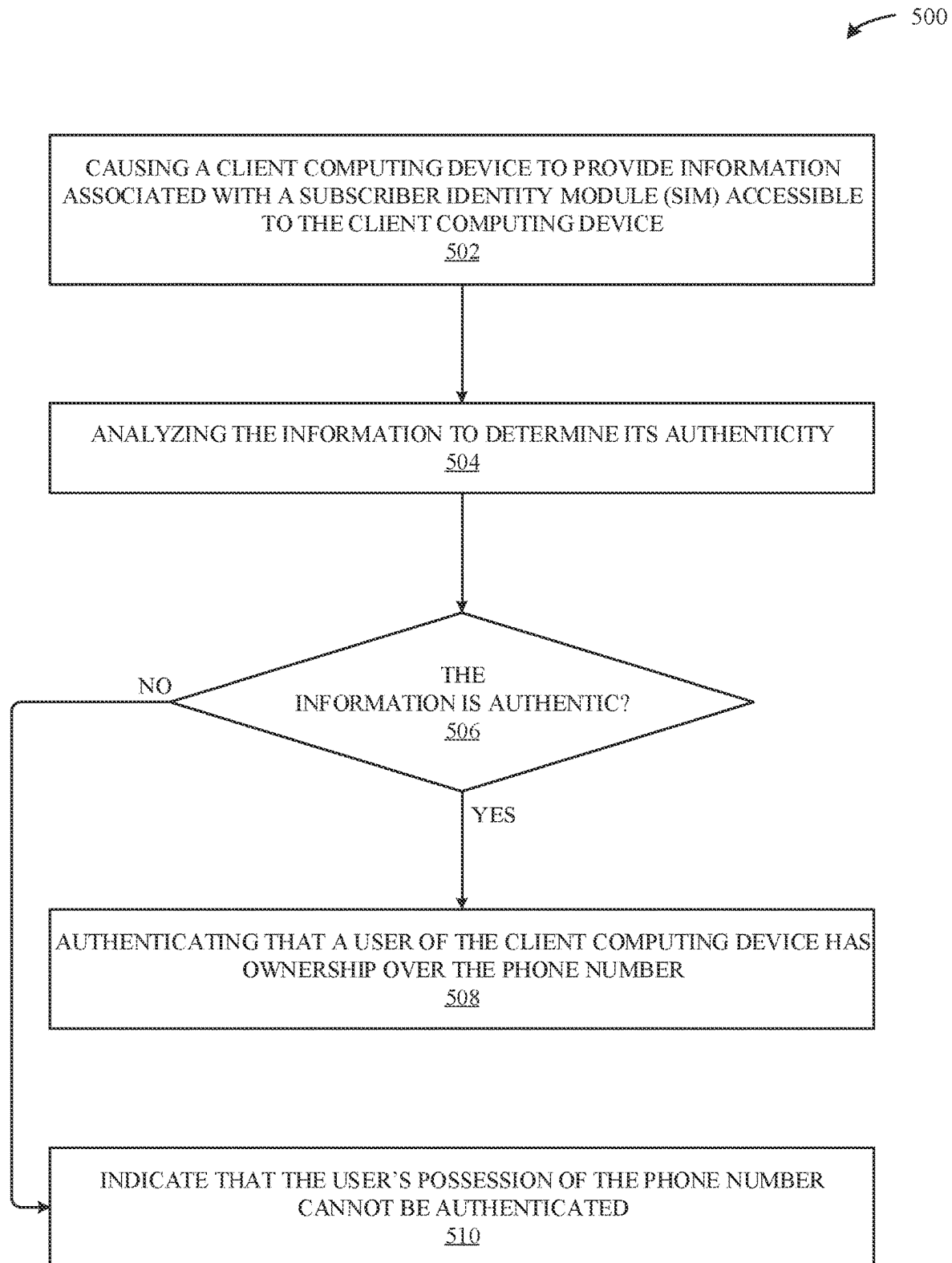


FIG. 5

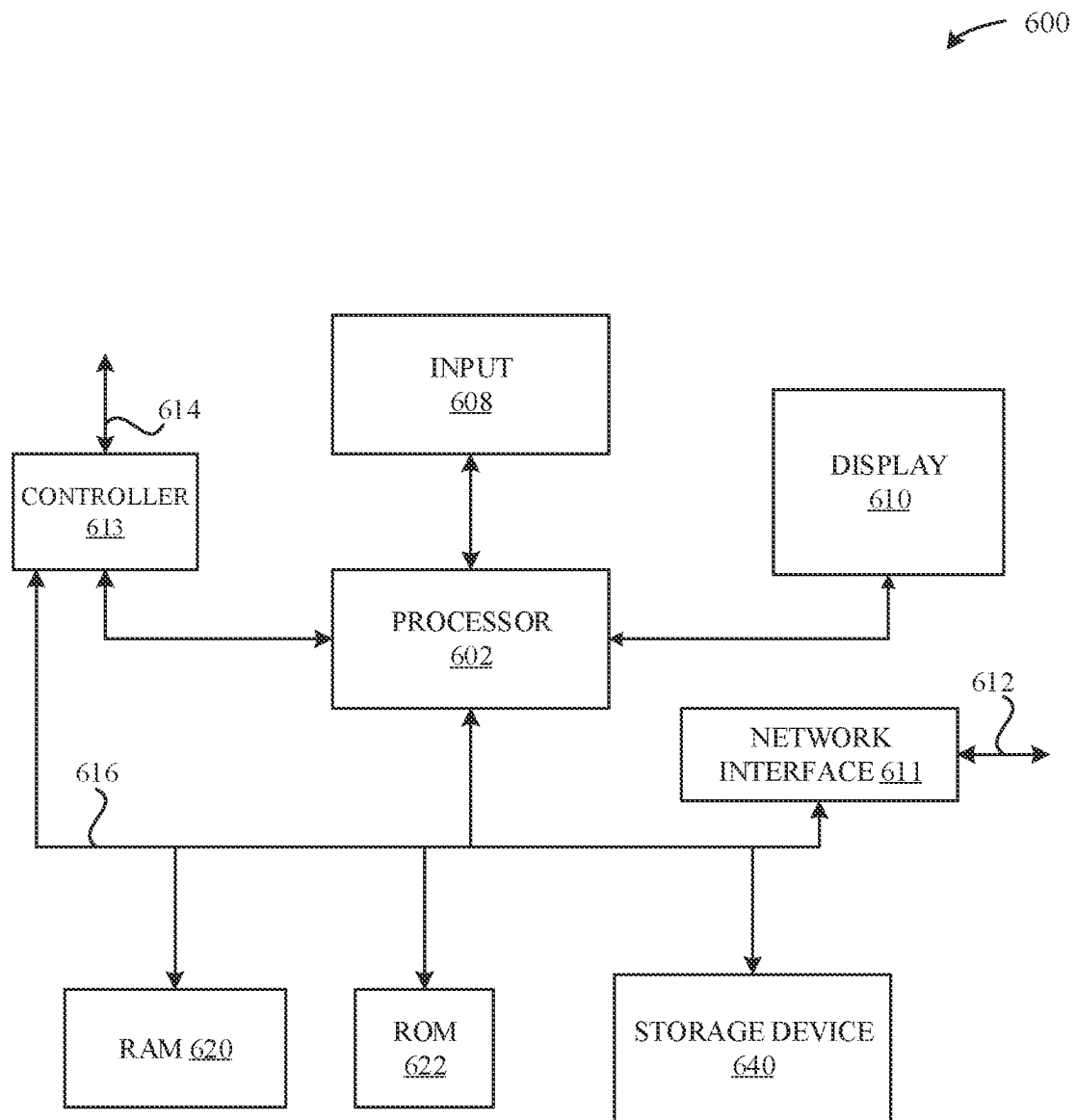


FIG. 6

1

TECHNIQUES FOR IMPLEMENTING PHONE NUMBER-BASED USER ACCOUNTS WITH PERMISSIONS TO ACCESS VARYING LEVELS OF SERVICES UTILIZING VISIBLE AND HIDDEN CONTACT ADDRESSES

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 17/017,560, filed Sep. 10, 2020, entitled “TECHNIQUES FOR IMPLEMENTING PHONE NUMBER-BASED USER ACCOUNTS WITH PERMISSIONS TO ACCESS VARYING LEVELS OF SERVICES UTILIZING VISIBLE AND HIDDEN CONTACT ADDRESSES”, issued Aug. 1, 2023 as U.S. Pat. No. 11,716,426, which claims the benefit of U.S. Provisional Application No. 63/041,912, entitled “TECHNIQUES FOR MANAGING PHONE NUMBER-BASED USER ACCOUNTS”, filed Jun. 21, 2020, the contents of all of which are incorporated by reference herein in their entirety for all purposes.

FIELD

The described embodiments relate generally to managing phone number-based user accounts. More particularly, the described embodiments involve enabling a user to register for a user account that is based on a phone number of the user. Subsequently, hidden and/or visible contact addresses based on the phone number of the user can be assigned to the user account to enable the user to access varying levels of services.

BACKGROUND

A conventional approach for creating a user account for a service involves prompting a user to provide an alphanumeric username and a password. Oftentimes, the user's desired alphanumeric username is unavailable or invalid, such that the user is forced to provide alternative alphanumeric usernames until one is available and valid. As a result, and over time, the user must recall varying usernames for the different services with which the user is registered, which can be frustrating and difficult for the user to effectively manage. For example, it may be necessary for the user to frequently utilize a “Forgot User ID” hyperlink that kicks off a cumbersome process of verifying credentials merely to indicate to the user the forgotten alphanumeric username.

Notably, some services have, at least in part, alleviated this issue by enabling users to instead register for user accounts using their email addresses. Typically, a given service requires a given user to prove possession of the email address prior to creating a user account based on the email address, e.g., by emailing the user with a verification hyperlink, or by requiring the user to input a code that is provided in an email sent to the email address. This approach is beneficial in that users are almost guaranteed to be unencumbered with availability/validation issues when attempting to register for user accounts using their email addresses.

Importantly, while email-based usernames have improved some of the issues associated with alphanumeric-based usernames, such email-based usernames do not provide any relief for individuals who are seeking to create user accounts based on their phone number. This deficiency is common because many users do not have email addresses, whereas

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virtually all users have a phone number. As a result, such users are required to register for an email address that they have no interest in utilizing.

Accordingly, what is needed is an improved user account creation approach that alleviates the above-described issues associated with alphanumeric-based and email-based usernames.

SUMMARY

The representative embodiments set forth herein disclose various techniques for enabling a user to register for a user account that is based on a phone number of the user.

One embodiment sets forth a method for managing phone number-based user accounts. According to some embodiments, the method can be implemented at a server computing device, and include the steps of (1) receiving, from a first client computing device, a request to create a user account based on a phone number, (2) in response to authenticating that the first client computing device is associated with the phone number: creating the user account based on the phone number, creating a hidden contact address based on the phone number, and assigning the hidden contact address to the user account, and (3) in response to receiving a notification that a second client computing device is associated with the phone number: requiring the user account to be updated to be based on a different phone number, or assigning, to the user account, a visible contact address based on the phone number.

Other embodiments include a non-transitory computer readable storage medium configured to store instructions that, when executed by a processor included in a computing device, cause the computing device to carry out the various steps of any of the foregoing methods. Further embodiments include a computing device that is configured to carry out the various steps of any of the foregoing methods.

Other aspects and advantages of the invention will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the described embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements.

FIG. 1 illustrates a block diagram of different computing devices that can be configured to implement different aspects of the various techniques described herein, according to some embodiments.

FIGS. 2A-2D illustrate conceptual diagrams for establishing a user account at a client computing device, according to some embodiments.

FIGS. 3A-3B illustrate a method for creating and managing phone number-based user accounts, according to some embodiments.

FIG. 4 illustrates a method for a code-based technique to determine whether a user is in possession of a Subscriber Identity Module (SIM) that is associated with a phone number, according to some embodiments.

FIG. 5 illustrates a method for an inquiry-based technique to determine whether a user is in possession of a SIM that is associated with a phone number, according to some embodiments.

FIG. 6 illustrates a detailed view of a computing device that can be configured to implement the various techniques described herein, according to some embodiments.

DETAILED DESCRIPTION

Representative applications of methods and apparatus according to the present application are described in this section. These examples are being provided solely to add context and aid in the understanding of the described embodiments. It will thus be apparent to one skilled in the art that the described embodiments may be practiced without some or all of these specific details. In other instances, well known process steps have not been described in detail in order to avoid unnecessarily obscuring the described embodiments. Other applications are possible, such that the following examples should not be taken as limiting.

In the following detailed description, references are made to the accompanying drawings, which form a part of the description and in which are shown, by way of illustration, specific embodiments in accordance with the described embodiments. Although these embodiments are described in sufficient detail to enable one skilled in the art to practice the described embodiments, it is understood that these examples are not limiting; such that other embodiments may be used, and changes may be made without departing from the spirit and scope of the described embodiments.

FIG. 1 illustrates a block diagram 100 of one or more server computing devices 150 that can be configured to implement various aspects of the techniques described herein. According to some embodiments, each server computing device 150 can represent any form of computing device, e.g., a smartphone, a tablet, a wearable, a desktop computing device, a rack-mounted computing device, and so on. It is noted that the foregoing examples are not meant to be limiting. On the contrary, each server computing device 150 can represent any form of computing device without departing from the scope of this disclosure. Although not explicitly illustrated in FIG. 1, a given server computing device 150 can include at least one processor, at least one memory, and at least one storage device that can operate in conjunction to enable the server computing device 150 to implement the various techniques set forth in this disclosure.

According to some embodiments, and as shown in FIG. 1, the server computing device 150 can be configured to manage a collection of user accounts 152, where each user account 152 can include a phone number 154. According to some embodiments, the phone number 154 of a given user account 152 represents a username property for the user account 152 and is based on a phone number of the user. For example, the phone number 154 can represent a string value that stores a text representation of the user's phone number (e.g., "111-222-3333"). Alternatively, the phone number 154 can represent a numerical value that stores a numerical representation of the user's phone number (e.g., 111222333). It is noted that the foregoing examples are not meant to be limiting, and that any approach can be used to store the user's phone number without departing from the scope of this disclosure. Additionally, each user account 152 can include a password 156, which represents any type of value (e.g., string, number, etc.) for storing the user's password for the user account 152.

As shown in FIG. 1, each user account 152 can include a hidden contact address 158 that is based on the phone number 154. Establishing a hidden contact address 158 for a given user account 152 places the user account 152 into an "unexposed" state that enables any client computing devices

102 logged-in with the user account 152 to participate in a subset of services 170 that are implemented by or accessible to the server computing device 150. According to some embodiments, the subset of services 170 can omit any service 170 within the services 170 for which there is an external alternative that commonly utilizes phone numbers as a basis for addressing/routing communications. The subset of services 170 can include, for example, cloud storage services, photo sharing services, location services, and so on. Conversely, the subset of services can omit, for example, audio and/or video services, messaging services, and so on.

According to some embodiments, each user account 152 can also include a visible contact address 160 that can be based on the phone number 154—or be based on another form of contact associated with the user, e.g., another phone number owned by the user, an email address belonging to the user, and so on. According to some embodiments, establishing a visible contact address 160 for a given user account 152 places the user account 152 into an "exposed" state that enables any client computing devices 102 logged-in with the user account 152 to obtain a higher level of access to the services 170 in comparison to the subset of services 170 (described above) accessible to the hidden contact address 158. The additional services 170 can include, for example, services 170 for which there is an external alternative that commonly utilizes phone numbers as a basis for addressing/routing communications, e.g., audio and/or video services, messaging services, and so on.

It is noted that a more detailed explanation of hidden and visible contact addresses, and the manner in which they affect the services 170 available to the user account 152, is described below in greater detail in conjunction with FIGS. 2A-2D and 3A-3B.

According to some embodiments, the server computing device 150 can be configured to communicate with one or more client computing devices 102. Each client computing device 102 can represent any form of computing device, e.g., a smartphone, a tablet, a wearable, a desktop computing device, a rack-mounted computing device, and so on. It is noted that the foregoing examples are not meant to be limiting, and that each client computing device 102 can represent any form of computing device without departing from the scope of this disclosure. Again, and although not explicitly illustrated in FIG. 1, it should be understood that client computing device 102 can include at least one processor, at least one memory, and at least one storage device to enable the client computing device 102 to interface with the server computing device(s) 150 so that the various techniques set forth in this disclosure can be implemented.

As shown in FIG. 1, each client computing device 102 can manage a user account 110 that corresponds to one of the user accounts 152 managed by the server computing devices 150. The user account 110 can be established on a client computing device 102 by performing a new user account 152 registration process, or by providing credentials for an existing user account 152. A new user account 152 registration process can involve, for example, one or more server computing devices 150 creating a user account 152 for the user, as well as the client computing device 102 establishing a user account 110 that corresponds to the newly-created user account 152. Alternatively, the existing user account 152 login process can involve, for example, the user providing login credentials—e.g., a phone number 154 and a password 156 for an existing user account 152—to a server computing device 150. When the login credentials are validated, the client computing device 102 can establish a user account 110 that corresponds to the existing user

account **152**. It is noted that a more detailed explanation of the user account creation/login processes is described below in greater detail in conjunction with FIGS. 2A-2D and 3A-3B.

Additionally, and as illustrated in FIG. 1, each client computing device **102** can be configured to implement services **112** that are configured to interface with the services **170** provided by the server computing devices **150**. For example, the services **112** can represent client-side implementations (e.g., applications, daemons, etc.) that correspond to server-side implementations and enable users of the client computing devices **102** to access the services **170**. Each client computing device **102** can also be configured to locally-execute one or more applications **114**, e.g., photo applications, mail applications, contacts applications, and the like.

Additionally, each client computing device **102** can include one or more Subscriber Identity Modules (SIMs) **130**. According to some embodiments, one or more of the SIMs **130** can represent a physical/removable SIM card (e.g., a 2FF, 3FF, 4FF, etc. SIM card) that is inserted into a respective receiving bay included in the client computing device **102**. Additionally, one or more of the SIMs **130** can represent an electronic Universal Integrated Circuit Card (eUICC) capable of managing one or more electronic (i.e., virtual/non-physical) Subscriber Identity Modules (eSIMs). In any case, each SIM **130** typically is associated with a phone number that enables the client computing device **102** to access wireless services provided by a respective Mobile Network Operator (MNO) with which the SIM **130** is associated. According to some embodiments, and as described in greater detail herein, the SIMs **130** can enable users to prove that they have both initial and continuing ownership over phone numbers that are tied to the user accounts **152**.

Accordingly, FIG. 1 sets forth a high-level overview of the different hardware/software architectures that can be implemented by the server computing device **150** and the client computing device **102** to implement the various techniques described herein. It is noted that the embodiments described herein are not limited to a single server computing device **150** and client computing device **102** interacting with one another. On the contrary, any number of server computing devices **150** can be configured to implement, in whole or in part, the techniques set forth herein, and can be configured to communicate with any number of client computing devices **102**. Additionally, it is noted that the services **170** described herein can be implemented by the server computing devices **150** (in whole or in part) or by other computing devices not illustrated in FIG. 1 that are configured to implement one or more of the services **170** (in whole or in part) or other services not illustrated in FIG. 1. Next, FIGS. 2A-2D, 3A-3B, and 4-6—which are described in greater detail below—set forth example conceptual and method diagrams in which the server computing device **150** can manage phone number-based user accounts in accordance with the techniques set forth herein.

FIGS. 2A-2D illustrate conceptual diagrams for establishing a user account **152** by way of a client computing device **102**, according to some embodiments. It is noted that the user interfaces provided in FIGS. 2A-2D are not meant to be limiting in any way, and that any form of user interface—as well as any procedure—can be implemented to achieve the techniques set forth herein without departing from the scope of this disclosure.

FIG. 2A illustrates a conceptual diagram **200** of a first user interface for creating a new user account **152** with the server

computing device **150**. As shown in FIG. 2A, the first user interface can enable a phone number to be input and used as a basis for establishing the new user account **152**, such as “111-222-3333”. According to some embodiments, the phone number field can be automatically populated to include the phone number, e.g., by detecting SIM **130** information that is accessible to the client computing device **102**. As also shown in FIG. 2A, the first user interface can also enable a desired password for the new user account **152** to be input and verified. After the information is entered, the “Submit” button can be selected to transmit the phone number/password information to the server computing device **150**.

FIG. 2B illustrates a conceptual diagram **210** of a second user interface for verifying that the phone number provided by the client computing device **102** is one of which the client computing device **102** is currently in possession. This can involve, for example, the server computing device **150** transmitting a temporary code to the phone number, where proof of receipt functions as proof that the phone number is accessible to the client computing device **102**. Various techniques can be used to authenticate the phone number without departing from the scope of this disclosure, including the code and inquiry-based techniques described in greater detail below in conjunction with FIGS. 4-5.

FIG. 2C illustrates a conceptual diagram **220** of a third user interface that displays a confirmation of the creation of the new user account **152**. The confirmation can follow the creation of the new user account **152**, where the phone number “111-222-3333” is assigned to the phone number **154** of the new user account **152**, the password is assigned to the password **156** of the new user account **152**, and the phone number “111-222-3333” is assigned to the hidden contact address **158** of the new user account **152**. As described herein, the new user account **152**, at this juncture, is in an “unexposed” state that enables the client computing device **102** to participate in a subset of services that are implemented by or accessible to the services **170**. Again, the subset of services **170** can omit services for which there is an external alternative that commonly utilizes phone numbers as a basis for addressing/routing communications (e.g., audio and/or video services, messaging services, etc.).

According to some embodiments, the new user account **152**—despite being in an unexposed state—can be permitted to access the omitted services **170** in a limited manner when one or more conditions are satisfied. In particular, the new user account **152** can be permitted to transmit communications over the audio and/or video services, messaging services, and/or other services so long as the communications are addressed to other user accounts **152** with which the new user account **152** is authorized to communicate. Such authorized user accounts **152** can include, for example, a family group of user accounts **152** of which the new user account **152** and at least one other user account **152** are members. This approach can enable, for example, parents and children to communicate with one another using their respective phone numbers **154** while preventing the children from receiving communications from other user accounts **152** (via their respective phone numbers **154**) that do not belong to the group.

It is noted that groups of user accounts **152** are not limited to family groups, and that any group of user accounts **152** can be established without departing from the scope of this disclosure. For example, a group of user accounts **152** can be established for a business to enable the user accounts **152** to communicate between one another using their respective phone numbers **154** while preventing the user accounts **152**

from communicating with other user accounts **152** (via their respective phone numbers **154**) that do not belong to the group.

Returning now to FIG. 2C, it is noted that in many cases, the user may desire to utilize the phone number to participate in the omitted services **170**. Accordingly, the third user interface also enables the new user account **152** to transition to an “exposed” state that enables the client computing device **102** to participate in such omitted services **170**. In FIG. 2C, a hypothetical user selects “YES” to this prompt, which causes the server computing device **150** to create a visible contact address **160** based on the phone number and associate the visible contact address **160** with the new user account **152**. A confirmation of success of this procedure is illustrated in the conceptual diagram **230** of a fourth user interface of FIG. 2D that displays a confirmation of transitioning the user account **152** to the “exposed” state.

Additionally, it is noted that the client computing device **102** can take appropriate action to create/manage a counterpart user account **110** that complements the user account **152** managed by the server computing device **150** as the user account **152** creation/management processes are executed. For example, FIG. 2C can involve the client computing device **102** creating the counterpart user account **110** in an “unexposed” state, and FIG. 2D can involve the client computing device **102** updating the counterpart user account **110** to indicate that it is in an “exposed” state. This update can enable, for example, other user interfaces for registering with and participating in services **170** that are available only to “exposed” user accounts.

FIGS. 3A-3B illustrate a method **300** for creating and managing phone number-based user accounts, according to some embodiments. As shown in FIG. 3A, the method **300** begins at step **302**, where the server computing device **150** receives, from a first client computing device **102**, a request to create a user account **152** based on a phone number. According to some embodiments, the request can be provided by the first client computing device **102** in response to a user of the first client computing device **102** seeking access to one or more of the services **170**. For example, the user of the first client computing device **102** may desire to enable data stored on the first client computing device **102** to be backed up to cloud storage included in the services **170**. In another example, the user of the first client computing device **102** may desire to enable the first client computing device **102** to receive communications (e.g., video, audio, text, etc.) provided by a communications platform included in the services **170**. In yet another example, the user of the first client computing device **102** may desire to enable the first client computing device **102** to share photos with friends and family via a photo sharing service that is implemented by or accessible to the services **170**.

At step **304**, the server computing device **150** determines whether the first client computing device **102** is associated with the phone number. According to some embodiments, the server computing device **150** and the first client computing device **102** can engage in communications that effectively enable the server computing device **150** to verify that the phone number is accessible to the first client computing device **102** (and presumably owned by the user). A more detailed explanation of techniques that can be used to make such a determination are provided below in conjunction with FIGS. 4-5. If, at step **304**, the server computing device **150** determines that the client computing device is associated with the phone number, then the method **300** proceeds to step **306**. Otherwise, the method **300** can restart at step **302** or terminate altogether.

At step **306**, the server computing device **150** creates the user account **152** based on the phone number. According to some embodiments, creating the user account **152** can involve assigning the phone number to the phone number **154** of the user account **152**. Creating the user account **152** can also involve assigning a password—e.g., one provided by the user, one generated by the first client computing device **102**, etc.—to the password **156** of the user account **152**.

At step **308**, the server computing device **150** assigns, to the user account **152**, a hidden contact address **158** based on the phone number. According to some embodiments, the execution of step **308** places the user account **152** into an “unexposed” state that enables the first client computing device **102** to participate in a subset of services **170**. According to some embodiments, and as previously described herein, the subset of services can omit any service **170** within the services **170** for which there is an external alternative that commonly utilizes phone numbers as a basis for addressing/routing communications. Consider, for example, a messaging service implemented by the services **170**, where the messaging service enables users to message one another using their phone numbers. This messaging service is akin to the Short Message Service (SMS)/Multimedia Messaging Service (MMS) implemented by Mobile Network Operators (MNOs) in that they also enable users to message one another using their phone numbers. Also consider, for example, a voice communications service (e.g., a Voice over Internet Protocol) implemented by the services **170**, where the voice communications service enables users to conduct audio/video phone calls using their phone numbers. This voice communications service is akin to what is commonly offered by MNOs to its subscribers by utilizing their respective phone numbers.

Turning now to FIG. 3B, at optional step **309**, the server computing device **150** receives, from the first client computing device **102**, a request to create a visible contact address **160** based on the phone number, to effectively transition the user account **152** from an “unexposed” state to an “exposed” state. According to some embodiments, the request can be generated in conjunction with the first client computing device **102** seeking to access services **170** not included in the subset of services described above in conjunction with step **308** of FIG. 3A (e.g., services **170** for which there is an external alternative that commonly utilizes phone numbers as a basis for addressing/routing communications, e.g., audio and/or video services, messaging services, and so on).

At optional step **310**, the server computing device **150** determines whether the first client computing device **102** is associated with the phone number. It can be necessary to re-perform this determination because the first client computing device **102** may issue the request despite no longer having ownership over the phone number. Consider, for example, a scenario in which a user account **152** is created using the phone number that is associated with the first client computing device **102** (e.g., as described above in conjunction with steps **302-306** of FIG. 3A). Subsequently, the first client computing device **102** is associated with a new phone number—e.g., a swap to a new SIM card, a provision of a new eSIM, etc.—but the user account **152** remains assigned to the phone number. In this scenario, when an attempt is made to assign a visible contact address **160** to the user account **152** based on the phone number, the attempt should be denied, as it is not appropriate to direct communications—specifically, communications over services **170** for

which there is an external alternative—to the first client computing device **102** given it is no longer associated with the phone number.

Accordingly, if, at step **310**, the server computing device **150** determines that the first client computing device **102** is associated with the phone number, then the method **300** proceeds to step **311**, where the server computing device **150** assigns, to the user account **152**, a visible contact address **160** based on the phone number. Otherwise, the server computing device **150** denies the request, and the method **300** proceeds to step **312** of FIG. **3A**.

At step **312** of FIG. **3A**, the server computing device **150** receives a notification that a second client computing device **102** (i.e., one different from the first client computing device **102** described above in conjunction with FIGS. **3A-3B**) is associated with the phone number. More specifically, the notification indicates that the phone number is now accessible to—and presumably owned by the user of—the second client computing device **102**. This can occur, for example, when the second client computing device **102** is in possession of a SIM card assigned to the phone number, when the second client computing device **102** is provisioned with an eSIM that is assigned to the phone number, and so on. In any case, this notification represents a conflict that the server computing device **150** should seek to resolve.

At step **314**, the server computing device **150** determines whether the user associated with the first client computing device **102** still owns the phone number. In one example, the first client computing device **102** and the second client computing device **102** may be owned by the same user, where the user simply swapped a physical SIM card (associated with the phone number) from the first client computing device **102** to the second client computing device **102**, or where an eSIM (associated with the phone number) was re-provisioned from the first client computing device **102** to the second client computing device **102**. This typically occurs when a user is upgrading to the second client computing device **102** while retaining ownership of the first client computing device **102**. In an alternative example, the first and second client computing devices **102** are respectively-owned by different users, and the second client computing device **102** has been assigned the phone number that at one point was assigned to the first client computing device **102**. This can occur, for example, when the phone number is disassociated from the first client computing device **102**, is returned to a pool of available phone numbers, and is subsequently assigned to the second client computing device **102**.

In any case, at step **314**, the server computing device **150** determines whether the user of the first client computing device **102** still owns the phone number. Again, a more detailed explanation of techniques that can be used to make such a determination are provided below in conjunction with FIGS. **4-5**. If, at step **314**, the server computing device **150** determines that the user of the first client computing device **102** no longer owns the phone number, then the method proceeds to step **316**.

At step **316**, the server computing device **150** updates the user account **152** to be based on a different phone number. According to some embodiments, step **316** can involve the server computing device **150** eliminating any visible contact addresses **160** of the user account **152**, and subsequently re-performing steps **302-311** described above in conjunction with FIGS. **3A-3B** to effectively assign the different phone number to the user account **152**. This can also involve causing the first client computing device **102**—and any other client computing devices **102** logged in with the user

account **152**—to update the user accounts **110** that correspond to the user account **152** such that the user accounts **110** are effectively assigned to the different phone number.

Alternatively, if, at step **314**, the server computing device **150** determines that the user of the first client computing device **102** does still own the phone number—i.e., the user owns at least one client computing device **102** that is assigned to the phone number—then the method **300** proceeds to optional step **318**. At optional step **318**, the server computing device **150** assigns, to the user account **152**, a visible contact address **160** that is based on the phone number. Such a configuration can be useful to the user in that the first client computing device **102** can be capable of participating in the services **170** normally not accessible to “unexposed” user accounts **152**—specifically, communications over platforms for which there is an external alternative—using the phone number originally assigned to the first client computing device **102**, notwithstanding the fact that the first client computing device **102** is no longer in possession of a SIM or an eSIM that is associated with the phone number. For example, the user can, using the above-described services **170**, receive voice/video phone calls, as well as text messages, addressed to the phone number at both the first and second client computing devices **102**.

FIG. **4** illustrates a method **400** for a code-based technique to determine whether a user is in possession of a SIM that is associated with a phone number, according to some embodiments. As shown in FIG. **4**, the method **400** begins at step **402**, where the server computing device **150** sends, to the phone number, an SMS message (e.g., a text message) that includes a temporary code. According to some embodiments, the client computing device **102** that includes the SIM can receive the text message via a Short Message Service (SMS) provided by a Mobile Network Operator (MNO) associated with the SIM.

At step **404**, the server computing device **150** receives a response code from the client computing device **102** that is associated with the phone number. According to some embodiments, the response code can be input by the user of the client computing device **102** into a prompt issued by a server computing device **150**, e.g., in association with the verification step **304** described above in conjunction with FIG. **3A**. Alternatively, the client computing device **102** can automatically identify the temporary code within the text message and provide it to the server computing device **150** to streamline the verification process.

At step **406**, the server computing device **150** determines whether the temporary code matches the response code. If, at step **406**, the server computing device **150** determines that the temporary code matches the response code, then the method **400** proceeds to step **408**, where the server computing device **150** authenticates that a user of the client computing device **102** possesses ownership of the phone number. Otherwise, the method **400** proceeds to step **410**, where the server computing device **150** indicates that the user’s possession of ownership of the phone number cannot be authenticated. This can involve, for example, causing the client computing device **102** to display a message indicating the authentication failure. According to some embodiments, the client computing device **102** can be permitted to re-attempt the authentication a threshold number of times, which can involve re-providing the temporary code or attempting to verify one or more updated temporary codes (by re-performing steps **402-406** described above).

FIG. **5** illustrates a method **500** for an inquiry-based technique to determine whether a user is in possession of a SIM that is associated with a phone number, according to

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some embodiments. As shown in FIG. 5, the method 500 begins at step 502, where the server computing device 150 causes a client computing device 102 to provide information associated with a Subscriber Identity Module (SIM) accessible to the client computing device 102, e.g., a particular SIM 130 accessible to the client computing device 102. According to some embodiments, the client computing device 102 can query the SIM 130 for credentials that prove the client computing device 102 is in possession of the SIM 130. For example, the SIM 130 can be associated with a public/private key pair, where the public key is known to the server computing device 150 (e.g., by way of a certificate authority), but the private key is known only to the SIM 130. In this example, the server computing device 150 can cause the client computing device 102/SIM 130 to respond to a random value challenge, which can involve encrypting the random value using the private key to produce an encrypted random value.

At step 504, the server computing device 150 analyzes the information to determine its authenticity. Continuing with the foregoing example, step 504 can involve the server computing device 150 utilizing the public key to decrypt the encrypted random value to produce—and subsequently verify—a decrypted random value.

At step 506, the server computing device 150 determines whether the information is authentic. Continuing with the foregoing example, step 506 can involve the server computing device 150 verifying that the decrypted random value matches the random value that was originally provided with the challenge. If, at step 506, the server computing device 150 determines that the information is authentic, then the method 500 proceeds to step 508, where the server computing device 150 authenticates that a user of the client computing device has ownership over the phone number. Otherwise, at step 510, the server computing device 150 indicates that the user's possession of the phone number cannot be authenticated. This can involve, for example, causing the client computing device 102 to display a message indicating the authentication failure. According to some embodiments, the client computing device 102 can be permitted to re-attempt the authentication a threshold number of times, which can involve re-attempting challenges (by re-performing steps 502-506 described above).

It is noted that the authentication techniques described above in conjunction with FIGS. 4-5 are not mutually exclusive, and that any combination of these techniques can be employed to authenticate ownership of a phone number without departing from the scope of this disclosure.

FIG. 6 illustrates a detailed view of a computing device 600 that can represent the different computing devices of FIG. 1 used to implement the various techniques described herein, according to some embodiments. For example, the detailed view illustrates various components that can be included in the client computing devices 102 and server computing devices 150 described above in conjunction with FIG. 1. As illustrated in FIG. 6, the computing device 600 can include at least one processor 602 that represents a microprocessor or controller for controlling the overall operation of the computing device 600. The computing device 600 can also include a user input device 608 that allows a user of the computing device 600 to interact with the computing device 600. For example, the user input device 608 can take a variety of forms, such as a button, keypad, dial, touch screen, audio input interface, visual/image capture input interface, input in the form of sensor data, and so on. Still further, the computing device 600 can include a display 610 that can be controlled by the at least

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one processor 602 (e.g., via a graphics component) to display information to the user. A data bus 616 can facilitate data transfer between at least one storage device 640, the at least one processor 602, and a controller 613. The controller 613 can be used to interface with and control different equipment through an equipment control bus 614. The computing device 600 can also include a network/bus interface 611 that couples to a data link 612. In the case of a wireless connection, the network/bus interface 611 can include a wireless transceiver.

As noted above, the computing device 600 also includes the at least one storage device 640, which can comprise a single disk or a collection of disks (e.g., hard drives). In some embodiments, the at least one storage device 640 can include flash memory, semiconductor (solid state) memory or the like. The computing device 600 can also include at least one Random-Access Memory (RAM) 620 and at least one Read-Only Memory (ROM) 622. The at least one ROM 622 can store programs, utilities, or processes to be executed in a non-volatile manner. The at least one RAM 620 can provide volatile data storage and can store instructions related to the operation of applications executing on the computing device 600.

The various aspects, embodiments, implementations or features of the described embodiments can be used separately or in any combination. Various aspects of the described embodiments can be implemented by software, hardware or a combination of hardware and software. The described embodiments can also be embodied as computer readable code on a computer readable medium for controlling manufacturing operations or as computer readable code on a computer readable medium for controlling a manufacturing line. The computer readable medium is any data storage device that can store data which can thereafter be read by a computer system. Examples of the computer readable medium include read-only memory, random-access memory, CD-ROMs, HDDs, DVDs, magnetic tape, and optical data storage devices. The computer readable medium can also be distributed over network-coupled computer systems so that the computer readable code is stored and executed in a distributed fashion.

The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of specific embodiments are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the described embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. A method for managing phone number-based user accounts, the method comprising, by a server computing device:

receiving a request to create a user account based on a phone number;
 authenticating an ownership of the phone number;
 creating hidden and visible contact addresses based on the phone number, wherein the hidden contact address permits access to a first subset of services that omits at least a second subset of services that are otherwise accessible using the visible contact address;

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assigning the hidden contact address to the user account and a first client computing device to permit the first client computing device to access the first subset of services; and

assigning the visible contact address to the user account and a second client computing device to permit the second client computing device to access the first and second subsets of services.

2. The method of claim 1, wherein authenticating the ownership of the phone number comprises:

receiving, from a client computing device that issues the request, a first verification that a Subscriber Identity Module (SIM) accessible to the client computing device is associated with the phone number,

receiving, from the client computing device, a second verification that the client computing device received a Short Message Service (SMS) code, or some combination thereof.

3. The method of claim 1, further comprising, prior to assigning the visible contact address to the user account: creating the visible contact address based on the phone number.

4. The method of claim 1, wherein the first subset of services includes:

client computing device recovery services, cloud sharing services, or some combination thereof.

5. The method of claim 1, wherein the second subset of services includes:

messaging-based communication services, audio/video-based communication services, or some combination thereof.

6. The method of claim 1, wherein:

the first client computing device comprises a first type of client computing device,

the second client computing device comprises a second type of client computing device, and

the first and second types of client computing devices are distinct from one another.

7. The method of claim 1, wherein creating the user account based on the phone number comprises assigning the phone number to the user account.

8. A non-transitory computer readable storage medium configured to store instructions that, when executed by at least one processor included in a server computing device, cause the server computing device to manage phone number-based user accounts, by carrying out steps that include:

receiving a request to create a user account based on a phone number;

authenticating an ownership of the phone number;

creating hidden and visible contact addresses based on the phone number, wherein the hidden contact address permits access to a first subset of services that omits at least a second subset of services that are otherwise accessible using the visible contact address;

assigning the hidden contact address to the user account and a first client computing device to permit the first client computing device to access the first subset of services; and

assigning the visible contact address to the user account and a second client computing device to permit the second client computing device to access the first and second subsets of services.

9. The non-transitory computer readable storage medium of claim 8, wherein authenticating the ownership of the phone number comprises:

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receiving, from a client computing device that issues the request, a first verification that a Subscriber Identity Module (SIM) accessible to the client computing device is associated with the phone number,

receiving, from the client computing device, a second verification that the client computing device received a Short Message Service (SMS) code, or some combination thereof.

10. The non-transitory computer readable storage medium of claim 8, wherein the steps further include, prior to assigning the visible contact address to the user account: creating the visible contact address based on the phone number.

11. The non-transitory computer readable storage medium of claim 8, wherein the first subset of services includes:

client computing device recovery services, cloud sharing services, or some combination thereof.

12. The non-transitory computer readable storage medium of claim 8, wherein the second subset of services includes:

messaging-based communication services, audio/video-based communication services, or some combination thereof.

13. The non-transitory computer readable storage medium of claim 8, wherein:

the first client computing device comprises a first type of client computing device,

the second client computing device comprises a second type of client computing device, and

the first and second types of client computing devices are distinct from one another.

14. The non-transitory computer readable storage medium of claim 8, wherein creating the user account based on the phone number comprises assigning the phone number to the user account.

15. A server computing device configured to manage phone number-based user accounts, the server computing device comprising:

at least one processor; and

at least one memory storing instructions that, when executed by the at least one processor, cause the server computing device to carry out steps that include:

receiving a request to create a user account based on a phone number;

authenticating an ownership of the phone number;

creating hidden and visible contact addresses based on the phone number, wherein the hidden contact address permits access to a first subset of services that omits at least a second subset of services that are otherwise accessible using the visible contact address;

assigning the hidden contact address to the user account and a first client computing device to permit the first client computing device to access the first subset of services; and

assigning the visible contact address to the user account and a second client computing device to permit the second client computing device to access the first and second subsets of services.

16. The server computing device of claim 15, wherein authenticating the ownership of the phone number comprises:

receiving, from a client computing device that issues the request, a first verification that a Subscriber Identity Module (SIM) accessible to the client computing device is associated with the phone number,

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receiving, from the client computing device, a second verification that the client computing device received a Short Message Service (SMS) code, or some combination thereof.

17. The server computing device of claim 15, wherein the steps further include, prior to assigning the visible contact address to the user account:

creating the visible contact address based on the phone number.

18. The server computing device of claim 15, wherein the first subset of services includes:

client computing device recovery services,
cloud sharing services, or
some combination thereof.

19. The server computing device of claim 15, wherein the second subset of services includes:

messaging-based communication services,
audio/video-based communication services, or
some combination thereof.

20. The server computing device of claim 15, wherein: the first client computing device comprises a first type of client computing device,

the second client computing device comprises a second type of client computing device, and

the first and second types of client computing devices are distinct from one another.

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